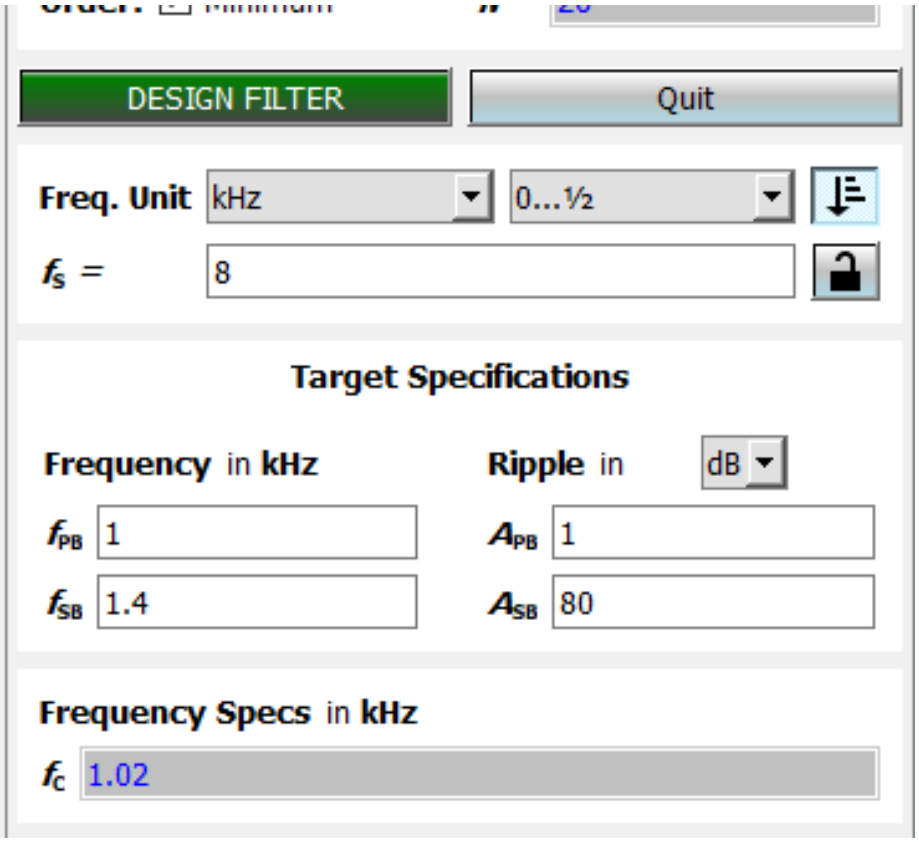
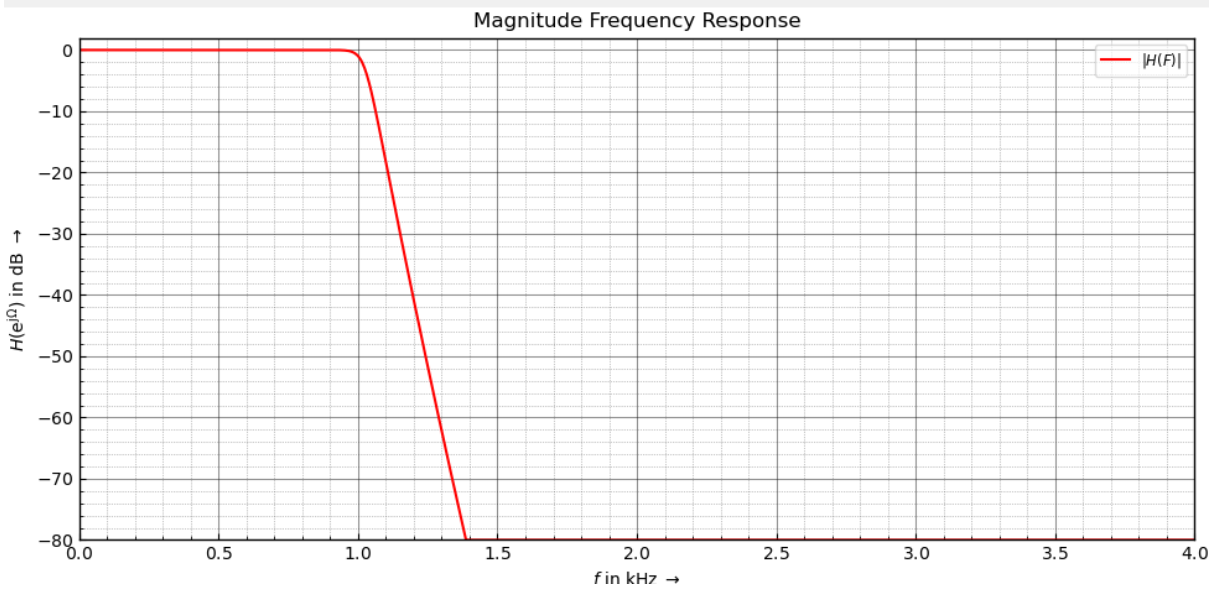
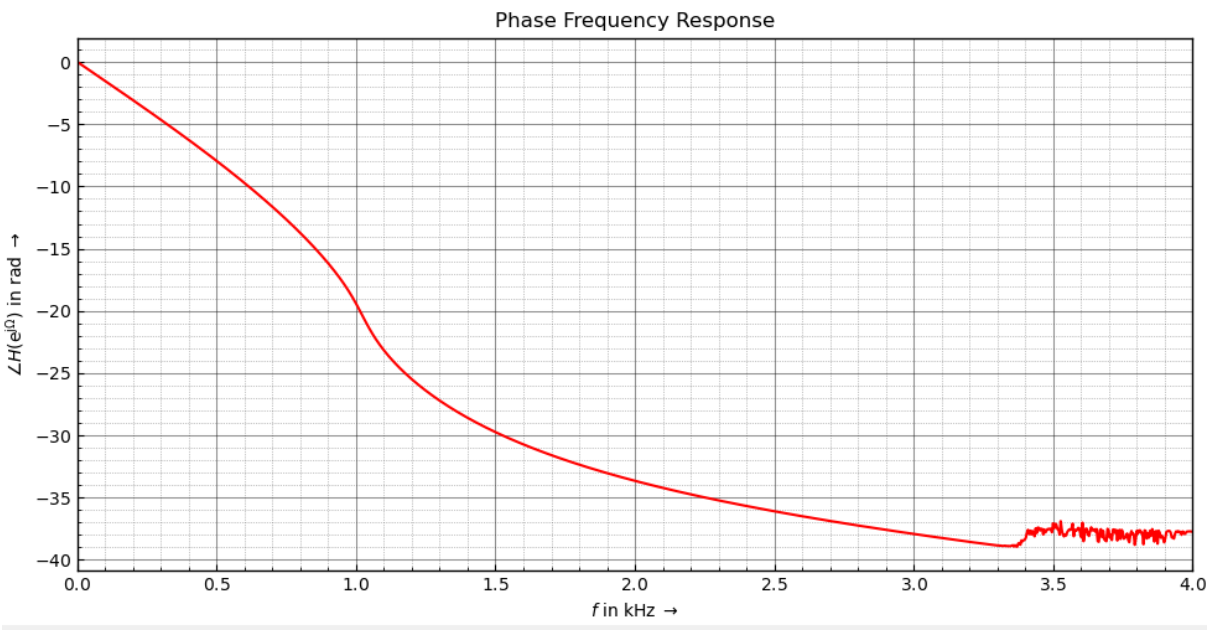
BOTSWANA INTERNATIONAL UNIVERSITY OF SCIENCE & TECHNOLOGY

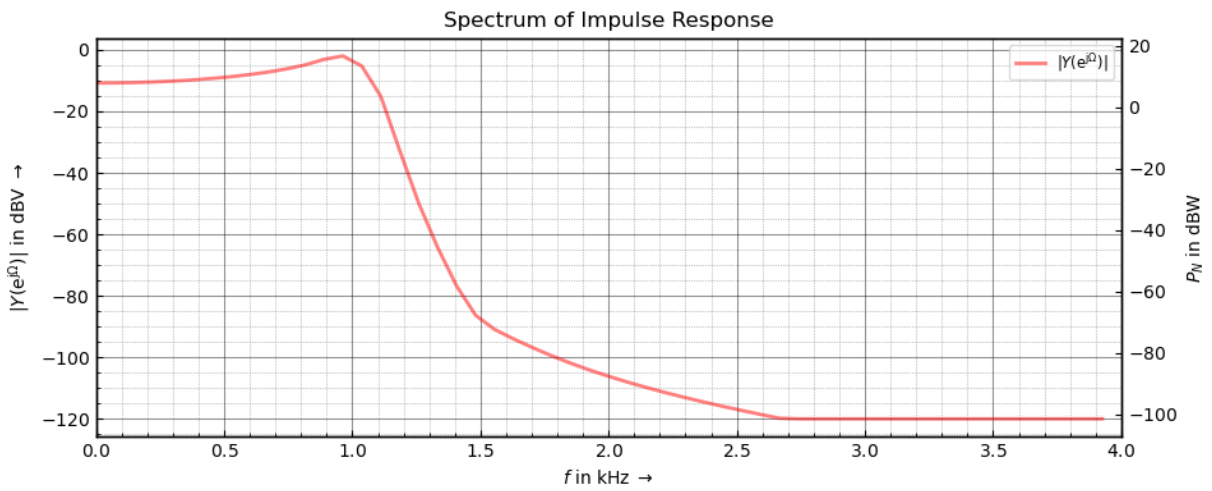
Simangaliso Mdluli 21000300

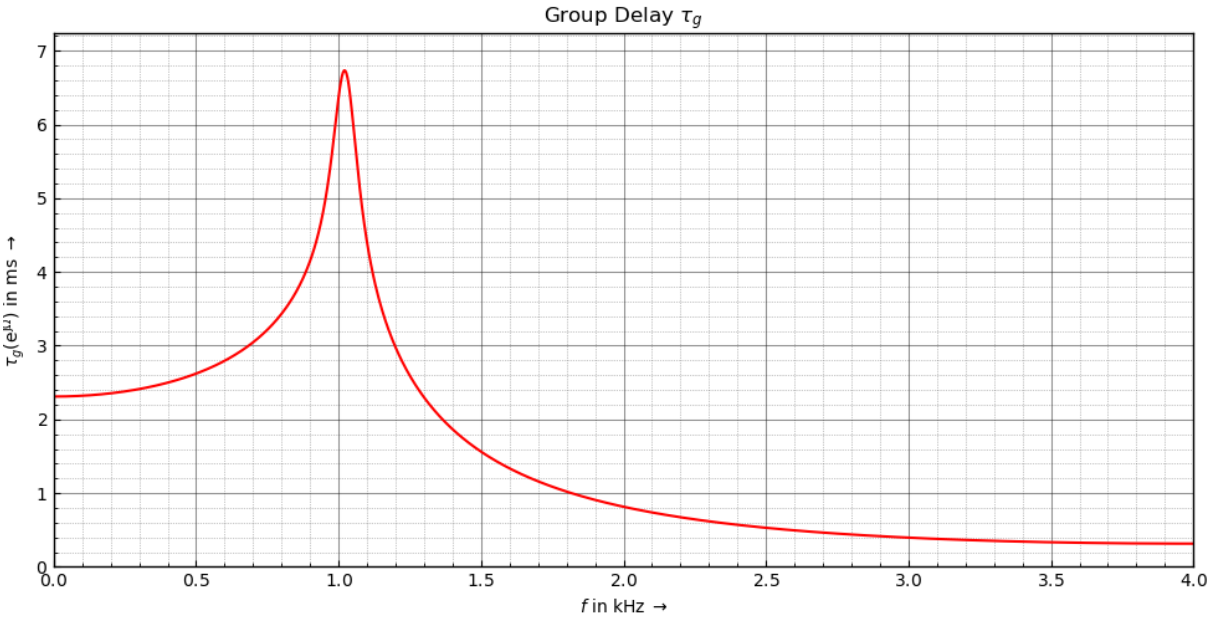
1) Butterworth filter

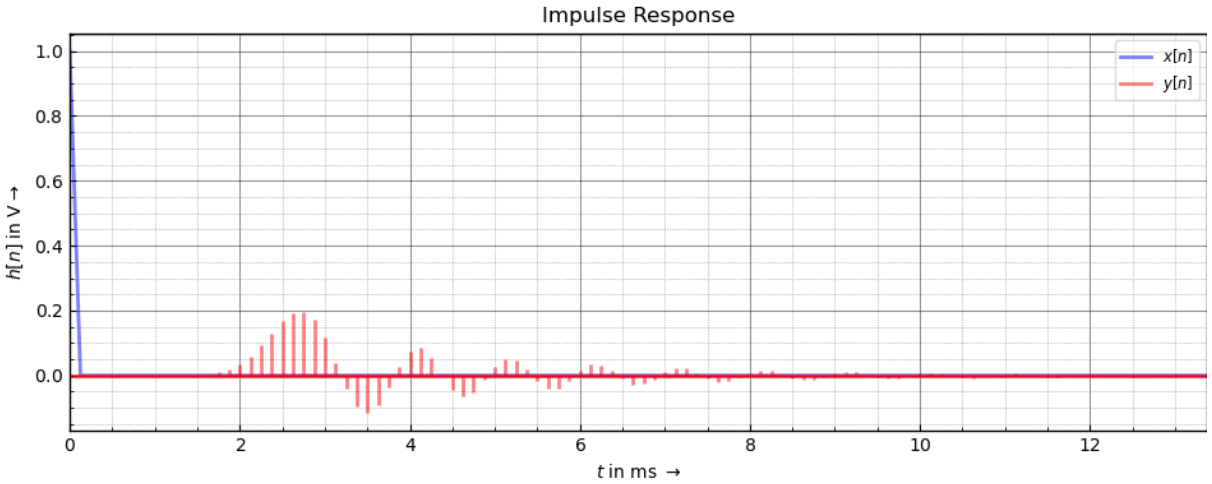


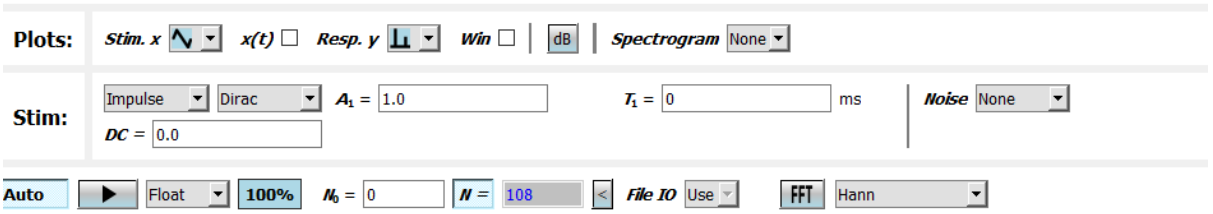


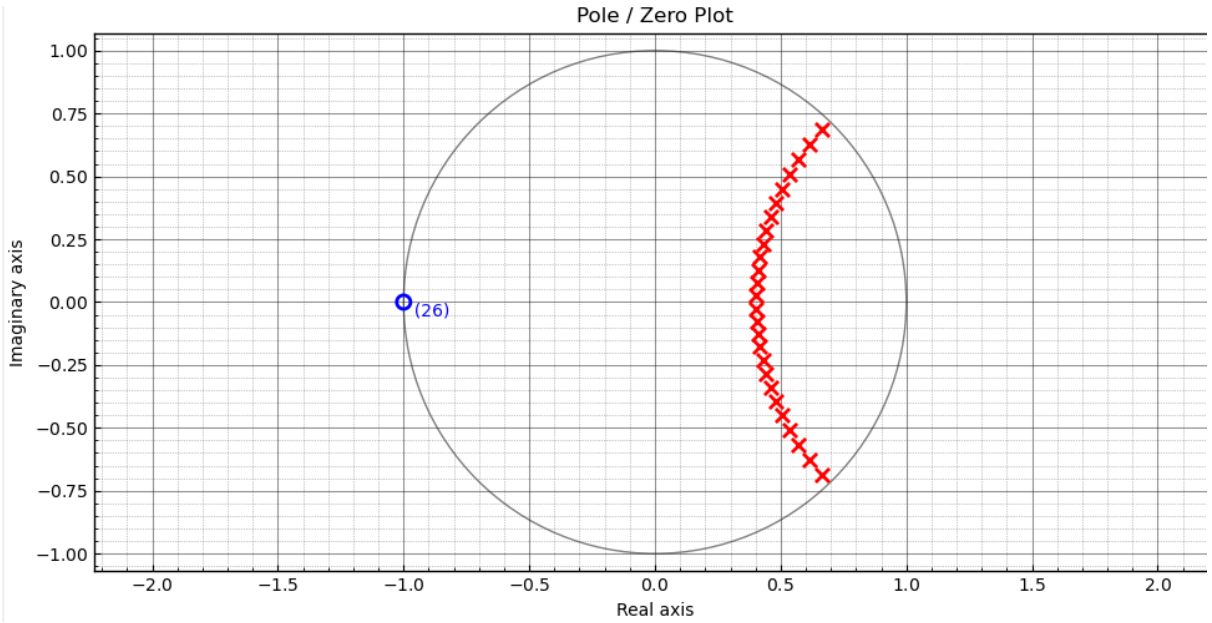

Phase Frequency Response
$\angle H(e^{j\Omega})$ in rad →
f in kHz →

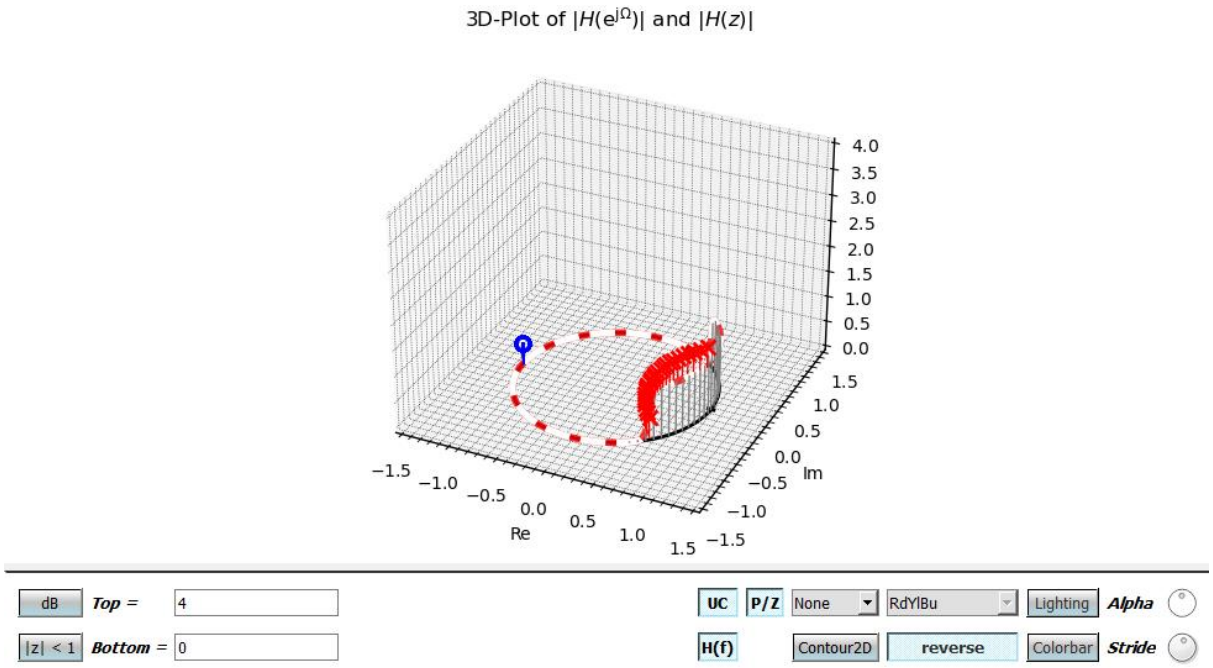

Spectrum of Impulse Response
$|Y(e^{j\Omega})|$ in dBV →
P_N in dBW
$|Y(e^{j\Omega})|$
f in kHz →


Group Delay τ_g
$\tau_g(e^{jf})$ in ms →
f in kHz →
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
1
2
3
4
5
6
7

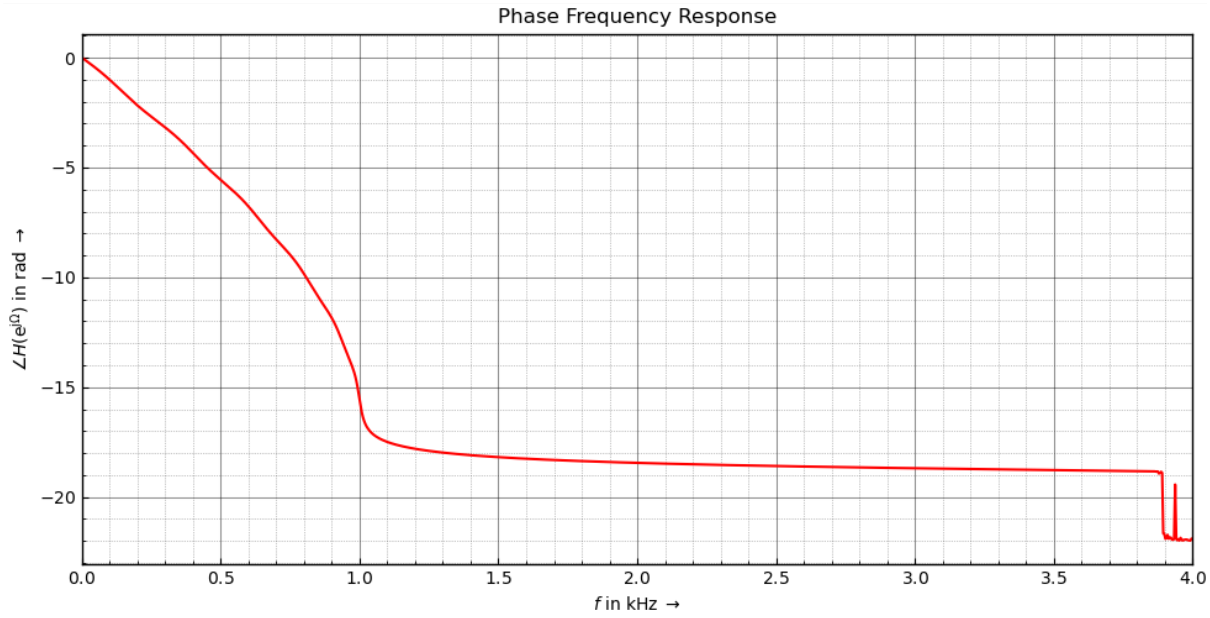

Impulse Response
$h[n]$ in V →
t in ms →
$x[n]$
$y[n]$
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
10
12

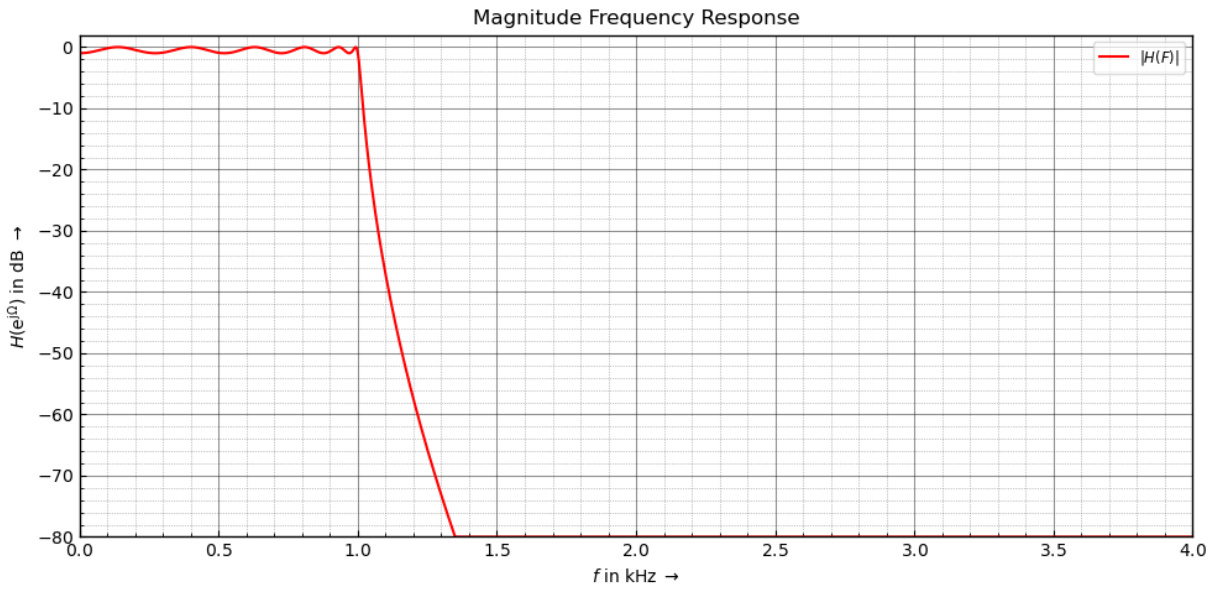

Plots:
Stim. x
x(t)
Resp. y
Win
dB
Spectrogram
None
Stim:
Impulse
Dirac
A_1 = 1.0
T_1 = 0
ms
Noise
None
DC = 0.0
Auto
Float
100%
N_0 = 0
N = 108
File IO
Use
FFT
Hann

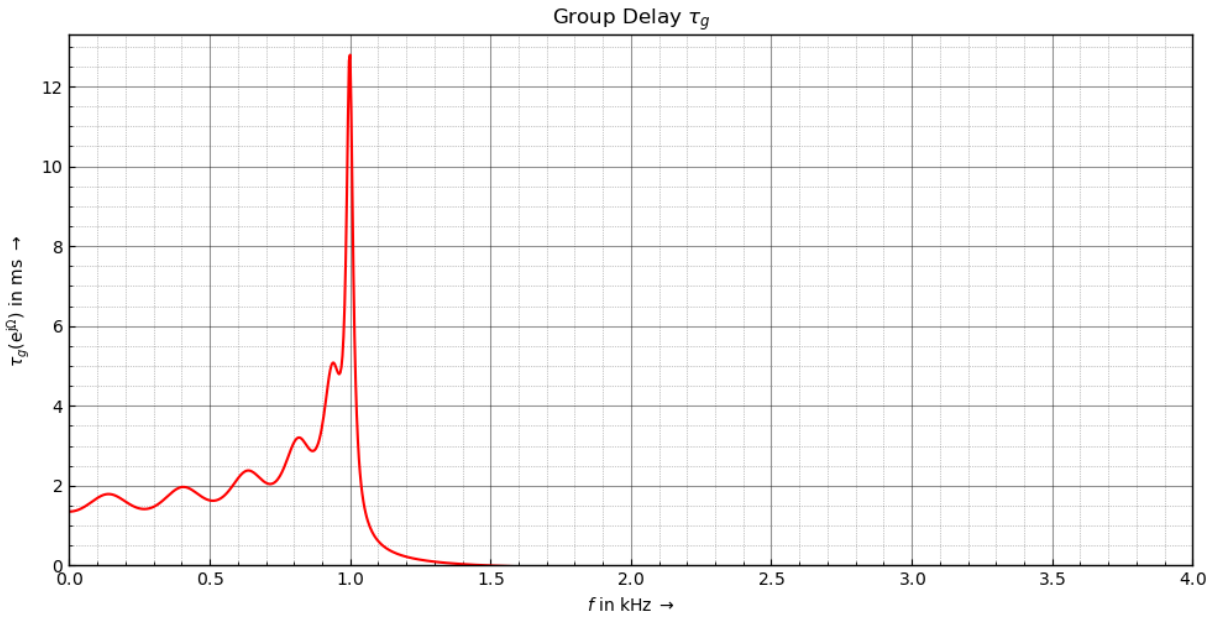

Pole / Zero Plot
Imaginary axis
Real axis
(26)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0

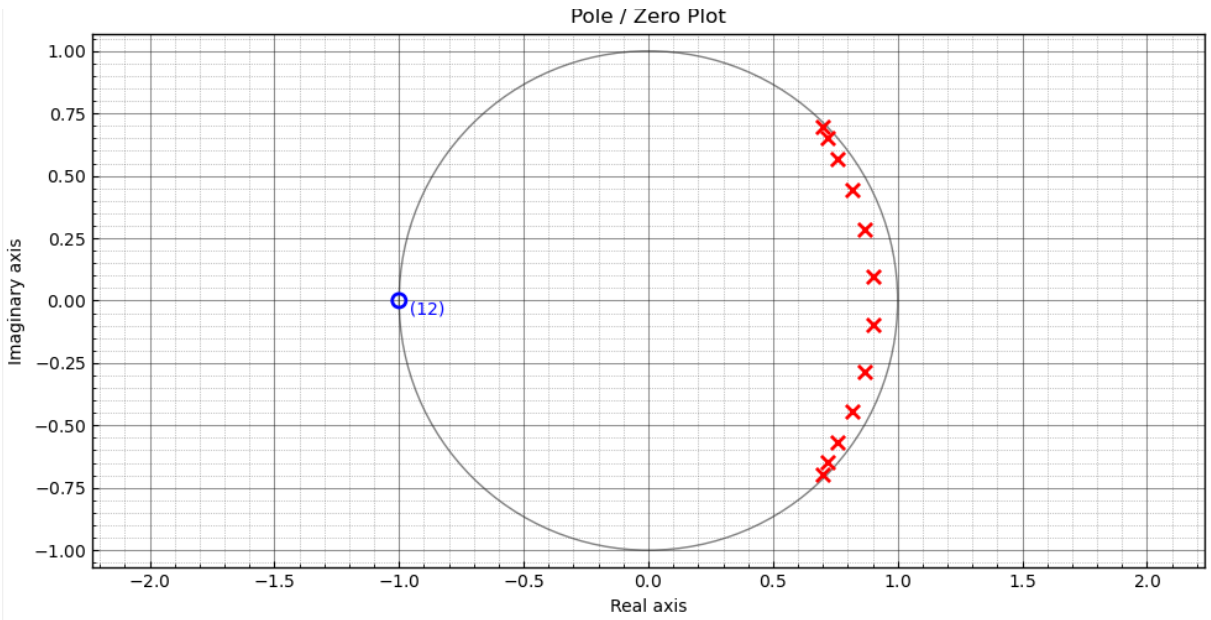

3D-Plot of |H(e^jΩ)| and |H(z)|
Re
Im
dB
Top =
4
|z| < 1
Bottom =
0
UC
P/Z
None
RdYlBu
Lighting
Alpha
H(f)
Contour2D
reverse
Colorbar
Stride

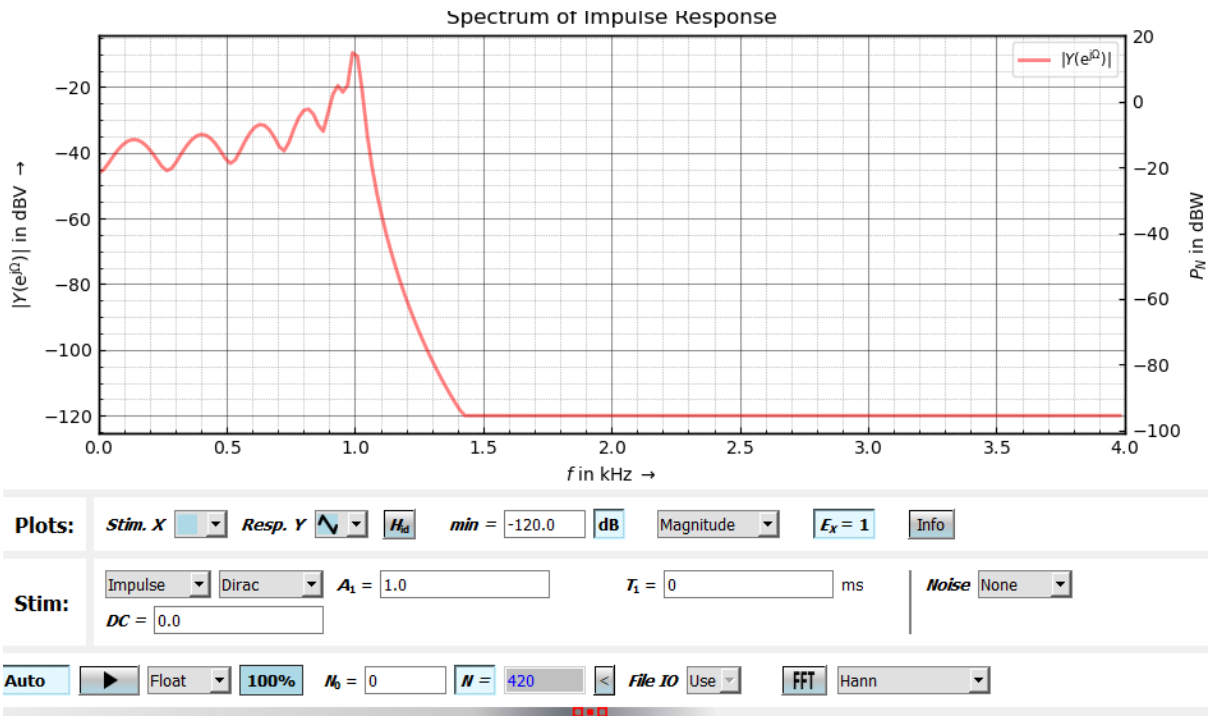
2) CHEBYSHEV

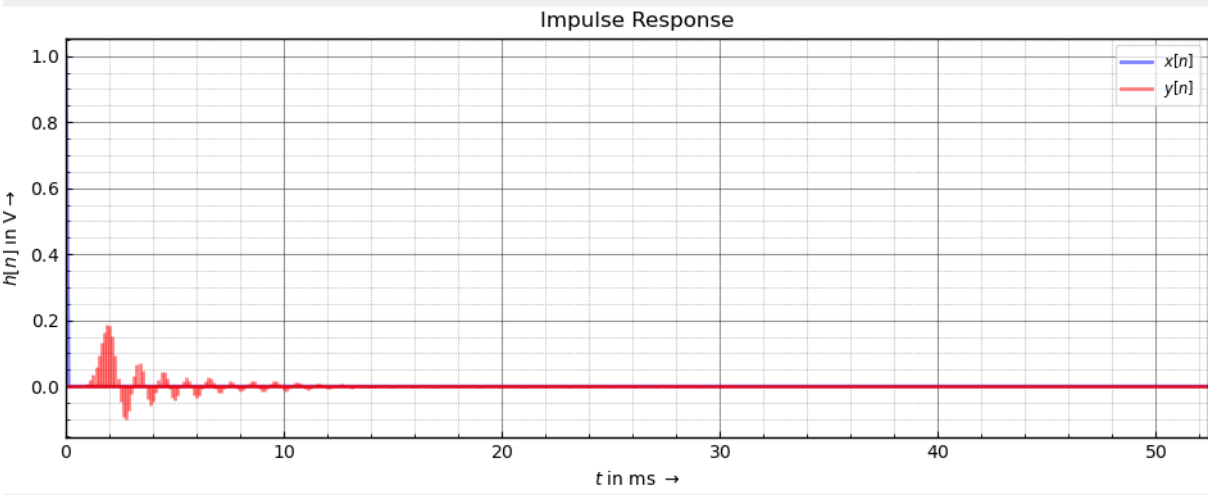

Phase Frequency Response
0
−5
−10
−15
−20
∠H(e^jΩ) in rad →
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
f in kHz →


Magnitude Frequency Response
|H(F)|
0
−10
−20
−30
−40
−50
−60
−70
−80
H(e^jΩ) in dB →
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
f in kHz →

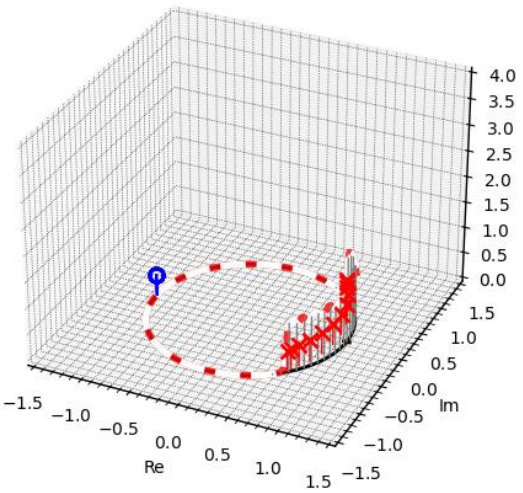

Group Delay τ_g
$\tau_g(e^{j\Omega})$ in ms →
f in kHz →
0
2
4
6
8
10
12
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0


Pole / Zero Plot
Imaginary axis
Real axis
(12)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0

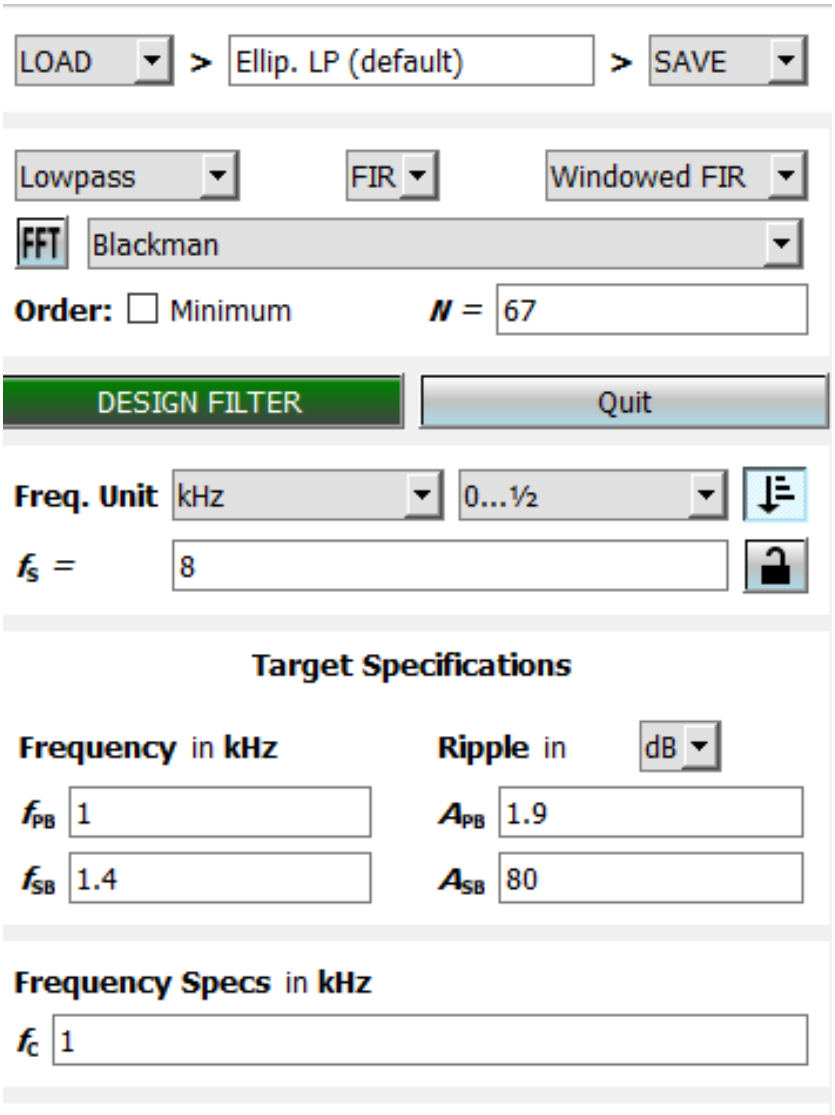

Spectrum of Impulse Response
|Y(e^jΩ)| in dBV →
P_N in dBW
f in kHz →
Plots:
Stim. X
Resp. Y
min = -120.0
dB
Magnitude
E_x = 1
Info
Stim:
Impulse
Dirac
A_1 = 1.0
T_1 = 0
ms
Noise
None
DC = 0.0
Auto
Float
100%
N_0 = 0
N = 420
File IO
Use
FFT
Hann

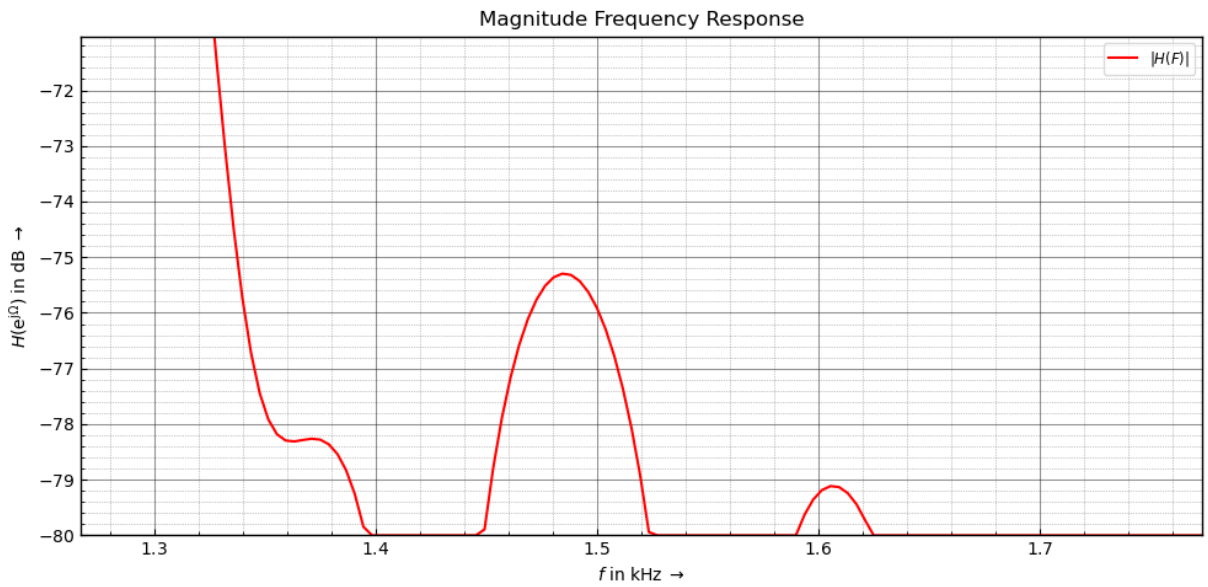

Impulse Response
x[n]
y[n]
h[n] in V →
t in ms →

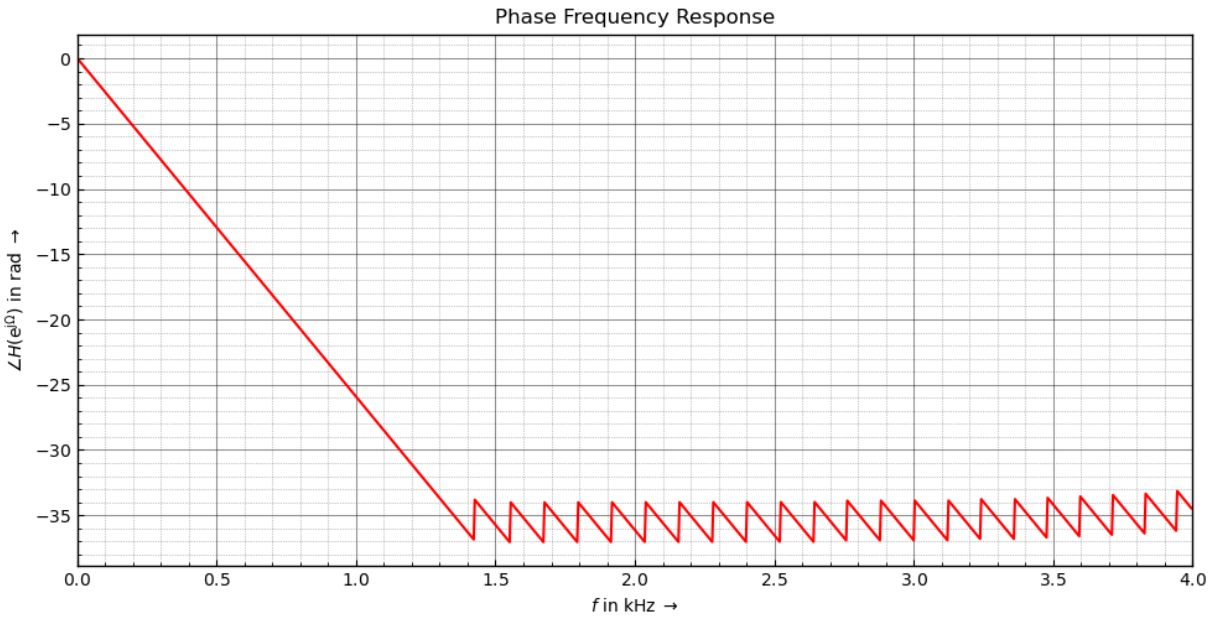
3D-Plot of $|H(e^{j\Omega})|$ and $|H(z)|$

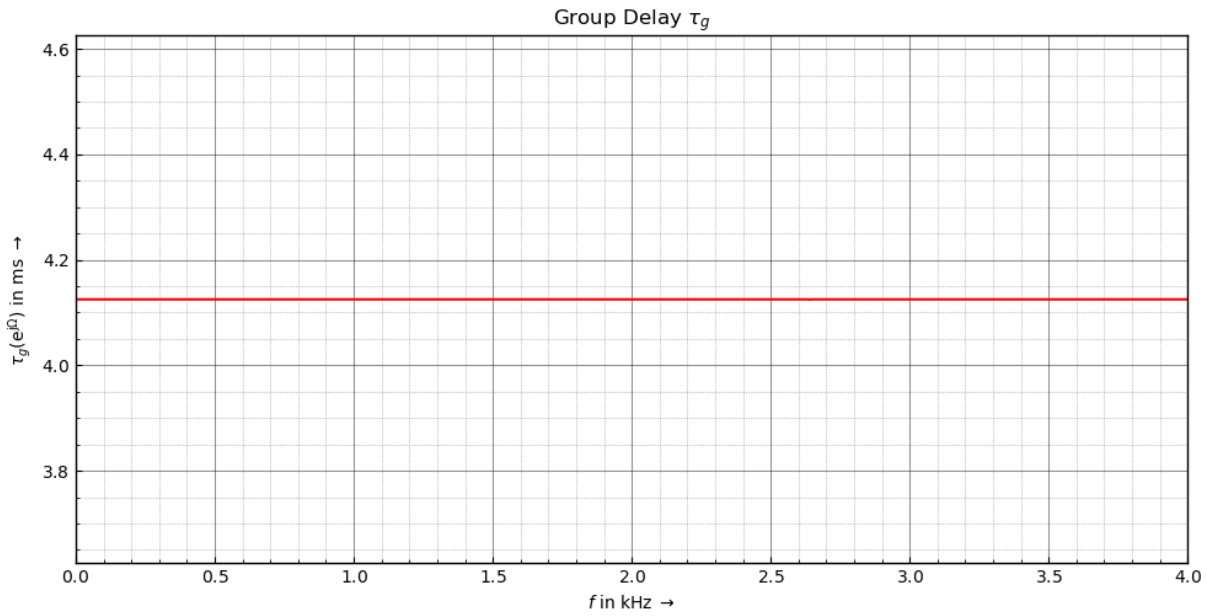


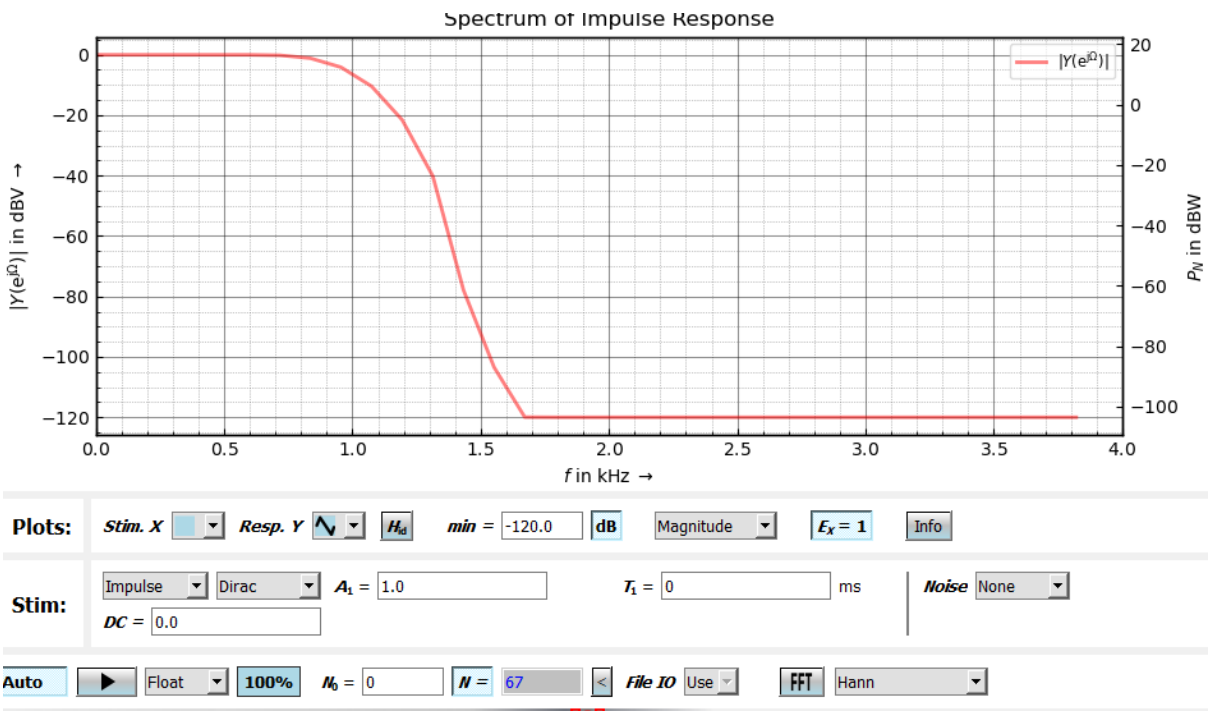
Q3 black window

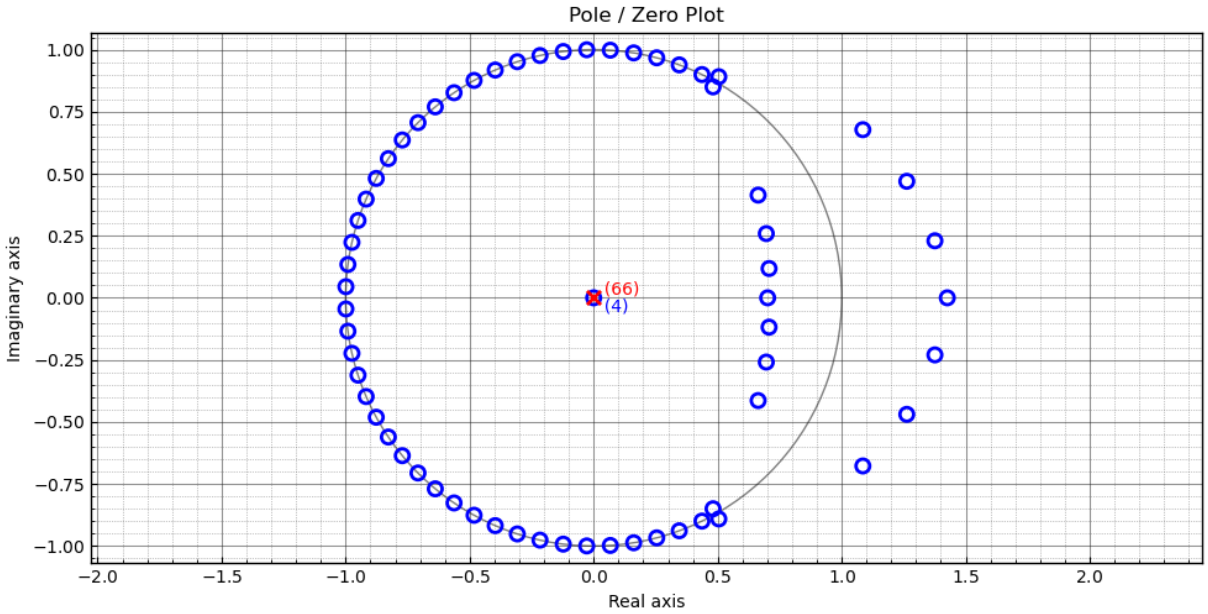


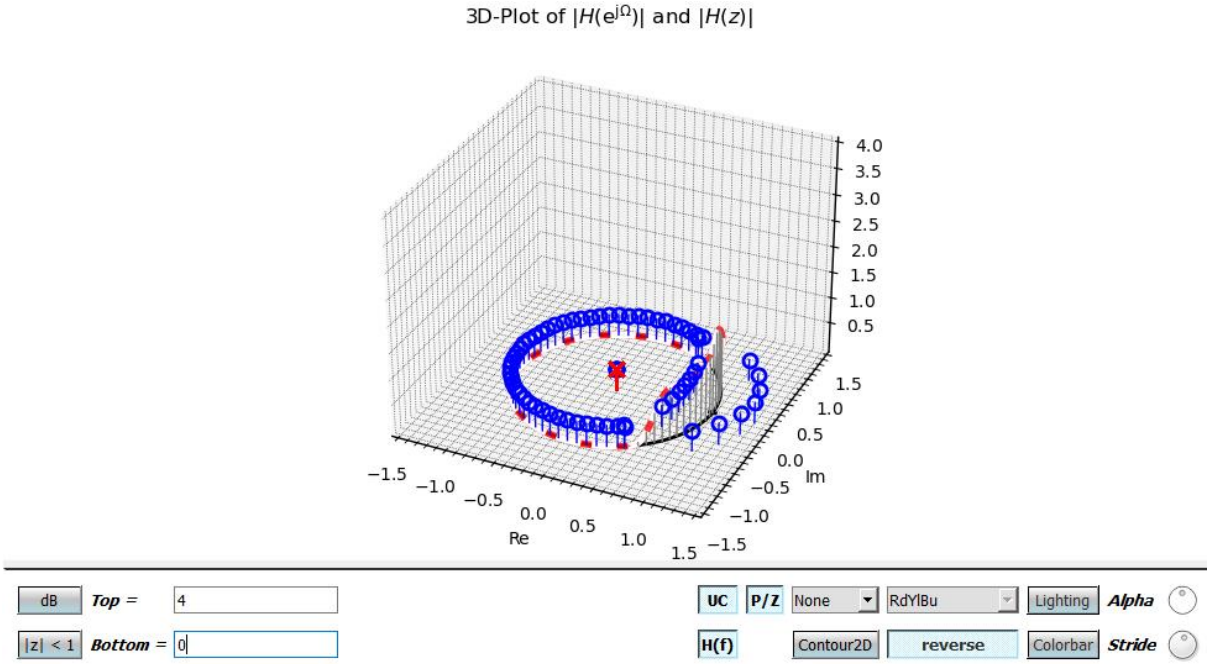

Magnitude Frequency Response
|H(F)|
−72
−73
−74
−75
−76
−77
−78
−79
−80
H(e^{jΩ}) in dB →
1.3
1.4
1.5
1.6
1.7
f in kHz →


Phase Frequency Response
0
−5
−10
−15
−20
−25
−30
−35
∠H(e^{jΩ}) in rad →
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
f in kHz →


Group Delay τ_g
$\tau_g(e^{j\Omega})$ in ms →
4.6
4.4
4.2
4.0
3.8
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
f in kHz →


Spectrum of Impulse Response
$|Y(e^{j\Omega})|$ in dBV →
P_N in dBW
$|Y(e^{j\Omega})|$
0
−20
−40
−60
−80
−100
−120
20
0
−20
−40
−60
−80
−100
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
f in kHz →
Plots:
Stim. X
Resp. Y
H_d
min = -120.0
dB
Magnitude
E_X = 1
Info
Stim:
Impulse
Dirac
A_1 = 1.0
T_1 = 0
ms
Noise
None
DC = 0.0
Auto
Float
100%
N_0 = 0
N = 67
File IO
Use
FFT
Hann


Pole / Zero Plot
Imaginary axis
Real axis
(66)
(4)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0


3D-Plot of |H(e^jΩ)| and |H(z)|
Re
Im
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
dB
Top =
4
|z| < 1
Bottom =
0
UC
P/Z
None
RdYlBu
Lighting
Alpha
H(f)
Contour2D
reverse
Colorbar
Stride

Discussion Questions

1. What is the order of the lowpass Butterworth lter you designed?

The order of the lowpass Butterworth filter designed in this lab is **26**. This order was chosen to achieve the desired frequency response while maintaining a balance between filter complexity and performance.

2. What is the order of the lowpass Chebyshev Type I lter you designed?

The order of the lowpass Chebyshev Type I filter designed in this lab is **12**. Chebyshev filters typically achieve a sharper roll-off compared to Butterworth filters with a lower order, which is reflected in the design.

3. Compare the memory usage of each of the 3 lters assuming a Direct Form II realization is employed. Do you see how ine cient the windowing technique is? How much more expensive in terms of memory is the windowing technique from the best IIR lter?

In a Direct Form II realization, memory usage is indicated by the number of adders required:

- **Butterworth IIR filter**: 52 adders.
- **Chebyshev IIR filter**: 24 adders.
- **FIR filter (Blackman window)**: 58 adders.

The windowing technique, specifically the Blackman window, uses the most memory with 58 adders. This highlights its inefficiency, as it requires more memory compared to both the Butterworth and Chebyshev filters. The Blackman filter achieves the required performance only with a very high order, leading to increased memory usage. In comparison, the Chebyshev filter is the most efficient, requiring the fewest number of adders. Thus, the windowing technique (Blackman FIR filter) is significantly more expensive in terms of memory than the best IIR filter (Chebyshev), which makes it less favorable for memory-constrained applications.